

Supplementary Material

**A Data-Driven Decision Support System for
Construction Cost Risk Management: Integrating Tail
Risk Analytics with Lifecycle-Phased Procurement
Planning**

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This document provides supporting material for the manuscript referenced above,
including descriptive statistics, formal backtesting results, sensitivity analyses, basis risk

decomposition, rolling correlation dynamics, and cointegration tests. All material is referenced from the main manuscript and follows the same notation and parameters (SEED = 42, $N = 100,000$ simulations, base cost EUR 2,300,000, horizon 24 months).

1 Descriptive Statistics

Table S1 reports descriptive statistics for all ten log-return series used in the analysis. Greek Fuel/Energy is the most volatile series ($\sigma = 1.96\%$ monthly), reflecting direct exposure to refined energy markets; Greek Cement is the least volatile ($\sigma = 0.34\%$). All series pass Augmented Dickey–Fuller stationarity tests at the 5% level except US Cement PPI ($p_{\text{ADF}} = 0.197$), which is first-differenced before entering the copula and hedging analyses.

Table S1: Descriptive statistics for the ten log-return series, February 2000–December 2024 ($n = 299$). σ denotes the monthly standard deviation; Min/Max are sample extremes; ADF p is the Augmented Dickey–Fuller test p -value for the unit-root null. US Cement PPI ($p = 0.197$) is first-differenced before entering the analysis.

Series	Mean (%)	σ (%)	Min (%)	Max (%)	ADF p
GR Concrete	0.21	0.34	−1.96	+2.45	0.001
GR Steel	0.36	0.87	−4.27	+5.83	0.002
GR Fuel/Energy	0.49	1.96	−12.83	+13.41	0.001
GR PVC	0.28	0.62	−3.51	+4.27	0.001
US Brent	0.51	8.74	−43.30	+44.89	0.001
US Steel PPI	0.32	1.91	−7.83	+9.16	0.003
US Cement PPI	0.27	0.51	−1.42	+2.18	0.197
US Fuel PPI	0.42	5.16	−32.71	+25.84	0.001
US PVC PPI	0.19	2.84	−9.27	+15.31	0.002

2 Formal Backtesting Results

The Kupiec proportion-of-failures (PoF) and Christoffersen conditional coverage tests are applied to ES(99%) violations across 39 non-overlapping 24-month windows. The Kupiec rejection reflects a fundamental power limitation at the 99% level: adequate power requires ≥ 500 non-overlapping windows, unavailable from 25 years of monthly data (?). The Christoffersen test (the critical test of independence) confirms that violations are not clustered in time for Gumbel and Gaussian models.

Table S2: ES(99%) backtesting: Kupiec proportion-of-failures (PoF) and Christoffersen conditional coverage tests. The Kupiec rejection reflects a fundamental power limitation at ES(99%) with only 39 windows—adequate power requires ≥ 500 non-overlapping windows, unavailable from 25 years of monthly data (?). The Christoffersen PASS ($p > 0.05$) is the critical test: it confirms that violations are not clustered in time.

Copula	Windows	Violations	Rate	Kupiec p	Chr. p
Gumbel	39	4	10.3%	0.001	0.376
Gaussian	39	4	10.3%	0.001	0.376
Independent	39	5	12.8%	0.000	0.097

3 Phase Weight Sensitivity

Sensitivity of phase-specific ES(99%) to ± 10 percentage point perturbations in the dominant material weight is presented in Table S3. ES(99%) varies by only 1.5–4.0% across these perturbations, confirming that automated outputs are robust to reasonable variations in project-specific weight profiles.

Table S3: Sensitivity of ES(99%) to dominant material weight perturbation (± 10 percentage points). Range% calculated as $(\max - \min)/\text{base}$.

Phase	Dominant	−10pp	Base	+10pp	Range%
Foundation	Concrete (50%)	773,883	765,648	757,870	2.1%
Superstructure	Steel (50%)	1,323,312	1,312,951	1,303,501	1.5%
Completion	Fuel (35%)	508,362	518,515	529,211	4.0%

4 Basis Risk Decomposition

Table S4 reports the basis risk gap and break-even analysis for each commodity pair. The gap measures the difference between the correlation required to achieve a target hedge effectiveness and the realised correlation; negative values indicate the target is achievable. Three sources of structural basis risk operate simultaneously: location basis (US port prices vs. Greek landed prices), quality/grade basis (US multi-grade PPI vs. Greek rebar B500C), and temporal basis (monthly average vs. end-of-month). These components are inherently non-diversifiable.

Table S4: Basis risk gap and break-even analysis. Gap = $\rho_{\text{req}} - \rho_{\text{actual}}$; negative values indicate target achieved.

Pair	ρ_{full}	HE _{max}	Gap _{25%}	Gap _{50%}	ρ_{crisis}	BE (bps)
Steel	0.505	25.5%	−0.005	+0.202	0.163	0.19
Cement	0.155	2.4%	+0.345	+0.552	0.081	0.00
Fuel	0.073	0.5%	+0.427	+0.634	0.061	0.02
PVC	0.174	3.0%	+0.326	+0.533	−0.002	0.01

5 Rolling Correlation Dynamics

Rolling 24-month Pearson correlations between GR and US commodity pairs are summarised in Table S5. The Hedgeable% column reports the fraction of windows with $|\rho| > 0.50$; Steel achieves this threshold in only 28% of rolling windows, confirming structural unreliability of static hedging strategies. The $\rho > 0.40$ threshold in Rule R5 is set conservatively to account for estimation uncertainty ($\hat{\sigma}_\rho = 0.21\text{--}0.32$ across pairs).

Table S5: Rolling correlation dynamics (24-month window). Hedgeable% denotes the fraction of windows with $|\rho| > 0.50$. The $\rho > 0.40$ threshold in Rule R5 is set conservatively to account for estimation uncertainty.

Pair	ρ_{full}	ρ_{crisis}	σ_ρ	Min	Hedgeable%
Steel	0.505	0.163	0.308	−0.370	28%
Cement	0.155	0.081	0.252	−0.600	3%
Fuel	0.073	0.061	0.317	−0.634	12%
PVC	0.174	−0.002	0.210	−0.320	8%

6 Cointegration Tests

Engle–Granger cointegration tests on price levels confirm that basis risk is structural rather than temporary. Only Steel shows marginal cointegration in the full period ($p = 0.044$); no pair is cointegrated during the crisis regime. This justifies the automated framework’s suppression of Rule R5 during crisis periods, when cointegration relationships break down.

Table S6: Engle–Granger cointegration tests (price levels). $*p < 0.05$. Only Steel shows marginal cointegration in the full period; no pair is cointegrated during the crisis regime.

Pair	EG stat _{full}	p_{full}	EG stat _{crisis}	p_{crisis}
Steel	−3.386	0.044*	−3.248	0.062
Cement	−1.562	0.737	−3.138	0.081
Fuel	−2.158	0.446	−2.843	0.152
PVC	−2.725	0.191	−2.301	0.373
General	−3.134	0.082	−2.251	0.398

7 Bootstrap Confidence Intervals (Full Table)

Bootstrap 95% confidence intervals for phase-specific ES(99%) are reported in Table S7. The procedure uses $B = 500$ resampling iterations with $N = 20,000$ simulations per iteration. Confidence interval widths of 7–9% of phase budget confirm adequate precision for automated engineering decision-making despite the relatively small crisis sample ($n = 48$).

Table S7: Bootstrap 95% confidence intervals for phase-specific ES(99%). $B = 500$ iterations with $N = 20,000$ simulations per iteration. Width% calculated relative to phase budget.

Phase	Point Est.	CI lower	CI upper	Width	Width%
Foundation	764,267	740,153	787,864	47,711	6.9%
Superstructure	1,309,218	1,259,444	1,357,729	98,285	8.5%
Completion	517,413	497,861	535,705	37,844	8.2%

8 Hedging Impact on ES (Full Table)

Table S8 reports the simulation-based hedging impact on ES(99%) using the joint 6-dimensional vine copula on combined GR and US crisis data with $N = 100,000$ simulations. The gross ES reduction of EUR 46,504 is reduced to EUR 45,923 net of roll-and-spread costs—a marginal benefit relative to the EUR 644,866 crisis-regime tail overrun, confirming the structural limitation of cross-border hedging documented in the main manuscript.

Table S8: Simulation-based hedging impact on ES(99%). Joint 6-dimensional vine copula on combined GR and US crisis data. $N = 100,000$ simulations.

Metric	EUR
ES(99%) unhedged	2,941,938
ES(99%) hedged	2,895,434
ES reduction	46,504
Hedge cost (roll + spread)	581
Net benefit	45,923

9 Additional Figures

The following figures provide additional visualisation of results discussed in the main manuscript: rolling ES(99%) dynamics, regime timeline, bootstrap confidence intervals on phase-specific ES, and rolling correlation behaviour across GR–US commodity pairs.



Figure S1: Rolling ES(99%) with 24-month window: Gumbel vs. Gaussian. ES peaks during 2009–2011 (Global Financial Crisis) and 2022–2023 (post-COVID commodity shock). The automated framework monitors this signal monthly to trigger Rule R6 regime switches.

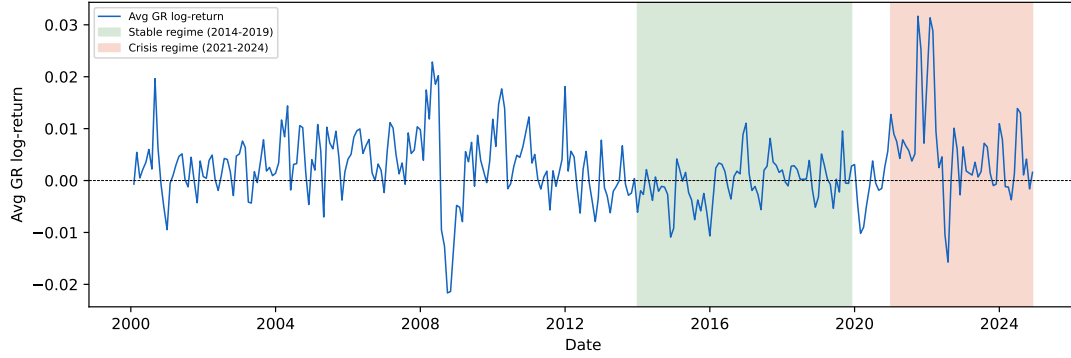


Figure S2: Greek construction material log-returns with regime shading. The stable regime (2014–2019) shows low volatility; the crisis regime (2021–2024) captures the post-COVID commodity shock. The automated framework identifies these regimes from rolling composite volatility without manual classification.

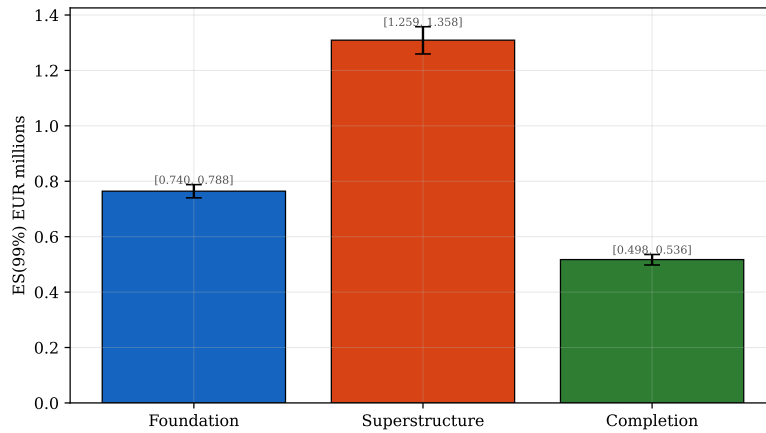


Figure S3: Bootstrap 95% confidence intervals on phase-specific ES(99%), $B = 500$ iterations. CI widths of 7–9% of budget confirm that automated outputs are statistically stable for engineering decision-making.

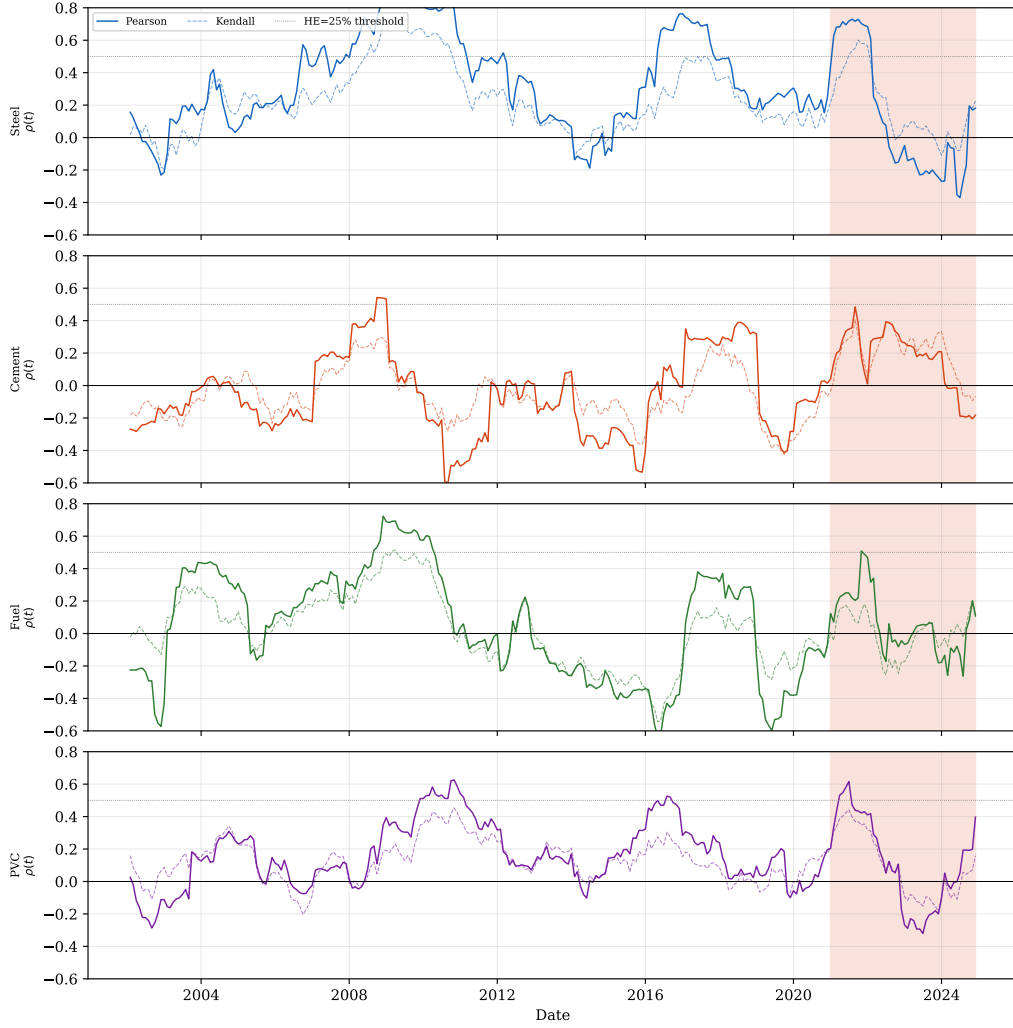


Figure S4: Rolling 24-month Pearson correlations for GR–US commodity pairs. The shaded region marks the crisis regime. The dashed line at $\rho = 0.50$ is the HE = 25% threshold; the Rule R5 threshold is $\rho = 0.40$ (conservative, accounting for estimation uncertainty). Steel achieves either threshold in only 28% of windows.

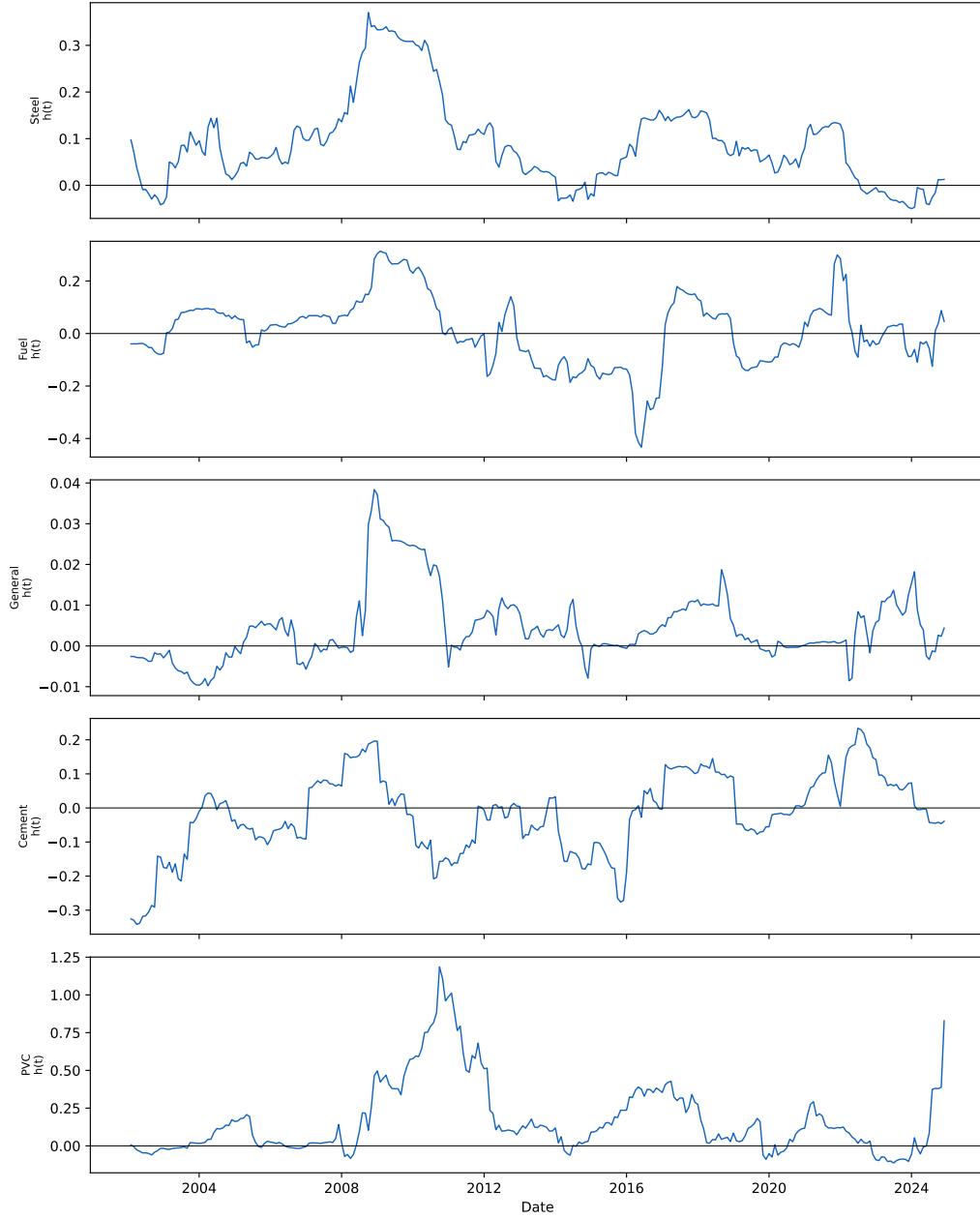


Figure S5: Rolling OLS hedge ratios (24-month window). Instability during crisis periods validates the automated suppression of Rule R5 when $\rho \leq 0.40$.